- Perform backtesting to evaluate model predictions against historical data.
- Compare models to select the one with the best trade-off between accuracy and complexity.

6. Deployment and Visualization:
    - Develop interactive dashboards using tools like Tableau, Power BI, or Dash.
        - Display metrics such as predicted sales, confidence intervals, and trends.
        - Allow filtering by time period, product category, or location.
    - Integrate the model into inventory or ERP systems for real-time decision-making.

7. Scalability and Optimization:
    - Use cloud platforms like AWS or Google Cloud for handling large datasets and deploying models at scale.
    - Optimize forecasting pipelines for real-time predictions.
    - Implement scheduled retraining with updated data to maintain accuracy.

Outcome:

The forecasting model provides actionable insights into future sales, enabling businesses to optimize inventory levels, plan revenue strategies, and respond proactively to market demands. It empowers data-driven decision-making to enhance operational efficiency and profitability.

Example Code:

```
import pandas as pd
from statsmodels.tsa.statespace.sarimax import SARIMAX
from sklearn.metrics import mean_absolute_error, mean_squared_error import matplotlib.pyplot as plt

# Step 1: Load Data
data = pd.read_csv('sales_data.csv', parse_dates=['date'], index_col='date') # Step 2: Visualize Data
```